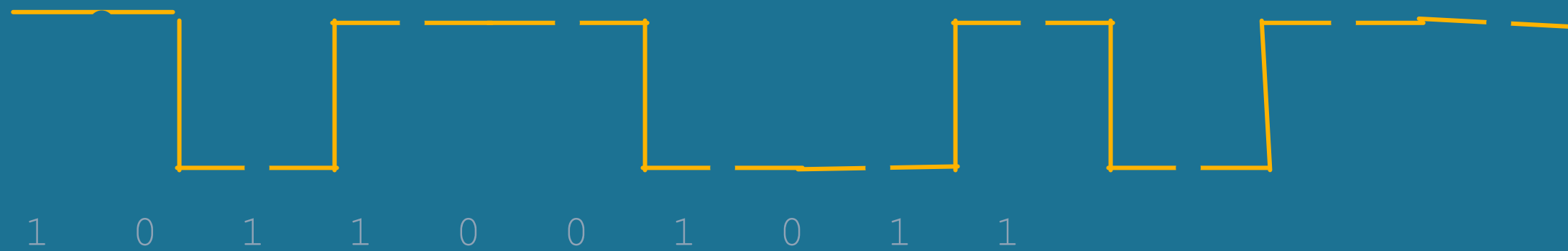


Optical Camera Communication

Video Message Decoding



video → bits → message

WEEK 1

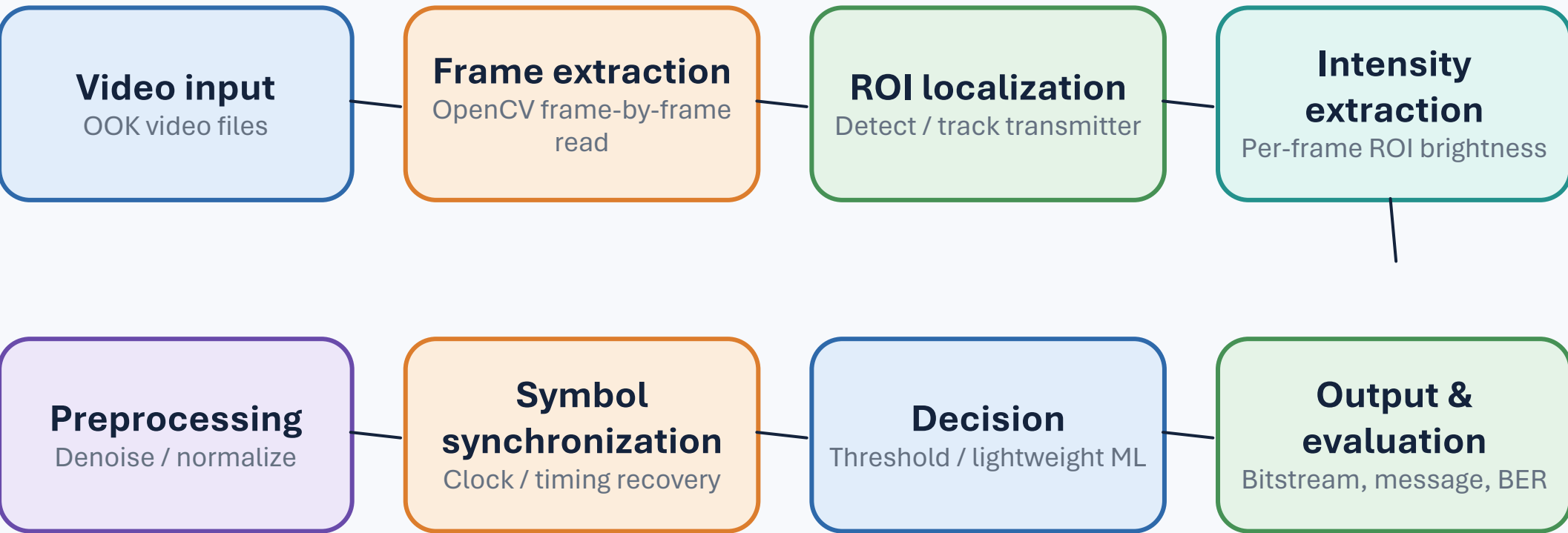
Optical Camera Communication (OCC) Video Message Decoding

Completed work

- 1 Problem framing
Restated the OCC decoding task: inputs, target output, and success criteria.
- 2 Domain review
Covered core CV, optical communication, signal-processing and ML concepts.
- 3 Literature study
Reviewed 10 closely related methods/systems and built a comparison table.
- 4 Pipeline & shortlist
Drafted the end-to-end software pipeline and shortlisted 3 candidate methods.

Pipeline drafted so far

Raw video → frames → transmitter ROI → intensity signal → preprocessing → synchronization → decision → decoded message



WEEK 1 — Findings & Blockers

2

What the literature tells us, and what remains unknown until the dataset arrives.

Key findings

- Prior OCC systems mostly assume synchronized or specialized modulation (UFSOOK, UPSOOK, rolling-shutter stripes). Our brief instead assumes generic OOK video with an unknown symbol/frame-rate relationship.
- Ambient light, auto-exposure and camera distance are dominant reported error sources — matching the planned sensitivity study.
- The closest real-world analog decodes ordinary, unsynchronized video end-to-end in software, supporting the proposed pipeline shape.
- CNN-based methods appear most useful for ROI localization and noisy-frame classification, rather than replacing signal-processing synchronization.

Open blockers

- 1 Dataset not yet received — frame rate, symbol rate, resolution and ground-truth labels are unconfirmed.
- 2 Unknown packet/framing structure — preamble, parity or other framing may sit above the raw bitstream.
- 3 ROI behaviour unclear — static per video or active tracking under camera/transmitter motion.
- 4 ML label granularity unconfirmed — per-bit/per-frame vs. per-message labels determine classifier feasibility.

LOOKING AHEAD

Dataset-driven execution plan and mentor decisions

01 Dataset inspection

02 Fixed-threshold baseline

03 Main signal-processing pipeline

04 ML-assisted — if labels permit

05 Sensitivity study